Statistical inferences for two samples

STAT 24510-30040

(Winter 2025)

Comparison of two samples is ubiquitous in scientific studies. This section walks through common methods for the comparison of two sample means.

Data: Two sets of observations or measurements: $\{x_1, \cdots, x_n\}$ and $\{y_1, \cdots, y_m\}$.

Question of interest: Did the two samples come from the same distribution, as opposed to, one sample having larger values than the other on average?

Assumptions:

$\{x_i, i = 1, \cdots, n\}$ are a sample from a population, the observations within the sample are independent.
$\{y_j, j = 1, \cdots, m\}$ are a sample from another population, the observations within the sample are independent.

Note the independence assumptions are <u>within</u> each sample.

Whether there is dependence <u>between</u> the two samples relies on the sampling method of the specific study.

To investigate general estimation methods, we assume

- $X_i, \cdots, X_n$ i.i.d. sampled from a distribution with mean μ_X, variance σ_X^2
- $Y_1, \cdots, Y_m$ i.i.d. sampled from a distribution with mean μ_Y, variance σ_Y^2

We are interested in comparing the two samples, often by comparing the two population means μ_X and μ_Y.

Standard formulations for two samples

- It is natural to consider $\bar{X} - \bar{Y}$ as basis for inference of $\mu_X - \mu_Y$.
- $E(\bar{X} - \bar{Y}) = \mu_X - \mu_Y$, the estimator is unbiased.
- To provide standard error to the above estimator, $Var(\bar{X} - \bar{Y})$ need to be estimated, contingent on sample properties.

Pairing - Correlated two samples

In many studies, the $i$the measurements in the two samples x_i and y_i actually are related, such as measurements 'pre' and 'after' a treatment from the same subject.

When $m = n$ and $Corr(X_i, Y_i) = \rho \not\approx 0$, very often $\rho > 0$. Then

$$\begin{aligned} Var(\bar{X} - \bar{Y}) &= Var(\bar{X}) + Var(\bar{Y}) - 2 \times Cov(\bar{X}, \bar{Y}) \\ &\leq Var(\bar{X}) + Var(\bar{Y}) \qquad\qquad when \quad \rho > 0. \end{aligned}$$

The inequality suggests that, to have a more precise variance estimate, it is appropriate to consider pairing X_i and Y_i by investigating their difference:

$$D_i = X_i - Y_i, \qquad i = 1, \cdots, n.$$

D_i's are i.i.d. under the i.i.d. assumption of each of the two samples, and

$$\bar{D} = \bar{X} - \bar{Y}$$

Let

$$\mu_D = E(D_i), \qquad \sigma_D^2 = Var(D_i)$$

be the common mean and common variance. They are related to the population parameters of the original two samples by

$$\mu_D = \mu_X - \mu_Y, \qquad \sigma_D^2 = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$$

Then essentially, we have changed the inference problem into that of a one-sample case, with n i.i.d. observations $D_1, \cdots, D_n$.

- An estimator for the variance of difference σ_D^2 is the sample variance.

  $$S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2 = \widehat{Var}(D_i) = \hat{\sigma}_D^2$$

  The sample variance is an unbiased estimator,

  $$E(S_D^2) = \sigma_D^2 = Var(D_i), \qquad E\left(\frac{S_D^2}{n}\right) = \frac{\sigma_D^2}{n} = Var(\bar{D}) = Var(\bar{X} - \bar{Y}).$$

- If $X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2)$, (by convention"$\sim$" indicate probability distribution of the random variable)

  $$\frac{\bar{D} - \mu_D}{\sigma_D/\sqrt{n}} \sim N(0,1), \qquad \frac{(n-1)S_D^2}{\sigma_D^2} \sim \chi_{n-1}^2.$$

  In addition, $\bar{D}$ and S_D^2 are independent. By the definition of t-distribution, the test statistic

  $$\frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$$

  is a pivot quantity, with a probability distribution not depending on the parameters.

  The pivot can be used to test hypothesis, for example, $H_o : \mu_D = 0$.

  A $(1-\alpha)\%$ confidence interval for the true difference $\mu_X - \mu_Y$ is

  $$(\bar{X} - \bar{Y}) \pm t_{n-1,1-\alpha/2}\frac{S_D}{\sqrt{n}}$$

  where the quantile $t_{n-1,1-\alpha/2}$ is defined by $\mathbb{P}(|t_{n-1}| \leq t_{n-1,1-\alpha/2}) = 1 - \alpha$.

- When the samples are not normally distributed, t_{n-1} can be used as an approximate distribution. For small or moderate n, often the t approximation usually is better than the normal approximation.

- When n is large, we may use the Central Limit Theorem,

  $$\frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} \sim \quad N(0,1) \qquad \text{approximately for large } n.$$

  The asymptotic pivotal property can be used to conduct hypothesis tests. A $(1-\alpha)100\%$ confidence interval for $\mu_X - \mu_Y$ can be approximated as

  $$(\bar{X} - \bar{Y}) \pm z_{1-\alpha/2}\frac{S_D}{\sqrt{n}}$$

  where the quantile z_* is defined as $\mathbb{P}(|Z| < z_{1-\alpha/2}) = 1 - \alpha, \ Z \sim N(0,1)$.

Pooling – Independent two samples with equal variance

Consider

$X_i, \cdots, X_n$ i.i.d. with mean μ_X, variance σ_X^2,
$Y_1, \cdots, Y_m$ i.i.d. with mean μ_Y, variance σ_Y^2.

The two samples $\{X_i, \cdots, X_n\}$ and $\{Y_1, \cdots, Y_m\}$ are independent.

Assume $\sigma_X^2 = \sigma_Y^2 = \sigma^2$. By independence and equal variance,

$$Var(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m} = \sigma^2\left(\frac{1}{n} + \frac{1}{m}\right)$$

Assume that we are interested in estimating $Var(\bar{X} - \bar{Y})$, by finding an appropriate estimator for σ^2.

- Both $S_X^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$ and $S_Y^2 = \frac{1}{m-1}\sum_{i=1}^m (Y_i - \bar{Y})^2$ are unbiased estimators of σ^2.
- A better unbiased estimator is the **pooled sample variance**

$$S_p^2 = S_{pooled}^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

  S_p^2 has smaller variance thus is more powerful than S_X^2, S_Y^2, that is, S_p^2 has larger power, where $Power = \mathbb{P}(\text{reject } H_o \mid H_a \text{ is true})$.

- If X_i, Y_i are of normal distributions, then the test statistic

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t_{n+m-2}$$

  Standard procedures can be applied utilizing the pivot to conduct hypothesis tests.

  A $(1-\alpha)\%$ confidence interval for the true difference $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm t_{n+m-2, 1-\alpha/2}\, S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

- When the samples are not normally distributed, t_{n+m-2} distribution can be used as an approximation.
- When <u>both</u> n and m are large, we may apply the Central Limit Theorem,

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \overset{approx.}{\sim} N(0,1)$$

  Hypothesis testing and confidence intervals can be constructed using the asymptotic pivot of $N(0,1)$.

- Pooling vs pairing — The importance of the independence assumption

  Consider the case $n = m$ with equal variance.
  Under assumption that the two samples are independent, the variance

$$Var(\bar{X} - \bar{Y}) = 2\sigma^2/n$$

  The pooled sample variance is the appropriate estimator to be used.

  If in fact, the two samples were correlations with not-small $corr(X_i, Y_i) = \rho > 0$, the variance

$$Var(\bar{X} - \bar{Y}) = 2\sigma^2(1-\rho)/n \ < \ 2\sigma^2/n$$

  In this case, the paired sample variance is the appropriate one to use.

  If the correlation is substantial and we fail to take it into consideration, the pooled sample variance estimator likely will overestimate the variance, and the estimate could be too large to be useful.

Independent two samples with unequal variance

Again consider $X_i, \cdots, X_n$ i.i.d. with mean μ_X, variance σ_X^2, independent of $Y_1, \cdots, Y_m$ i.i.d. with mean μ_Y, variance σ_Y^2.

Now $\sigma_X^2 \neq \sigma_Y^2$. By independence, the variance we are interested to estimate has the form

$$Var(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$$

which can not be simplified further (though reduced to the pooled form when sample sizes are equal).

- If $X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2)$, then $\bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right)$
- By the Central Limit Theorem for two samples, if <u>both</u> n and m are large,

$$\bar{X} - \bar{Y} \overset{approx.}{\sim} N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right)$$

- When the σ_X^2, σ_Y^2 are not given, $Var(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$ can be an estimated by unbiased estimator

$$\frac{S_X^2}{n} + \frac{S_Y^2}{m}, \qquad with \qquad E\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$$

A natural test statistic is

$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \tag{1}$$

<u>Remarks on test statistic (1)</u>

- When X_i, Y_i are of normal distributions,

$$(n-1)S_X^2/\sigma_X^2 \sim \chi_{n-1}^2, \qquad (m-1)S_Y^2/\sigma_Y^2 \sim \chi_{m-1}^2$$

  are of χ^2 distributions. However, since $\sigma_X^2 \neq \sigma_Y^2$,

$$\frac{S_X^2}{n} + \frac{S_Y^2}{m} \sim \frac{\sigma_X^2}{n(n-1)}\chi_{n-1}^2 + \frac{\sigma_Y^2}{m(m-1)}\chi_{m-1}^2$$

  This is not a multiple of a χ^2 distribution.

- T in (1) is not of t_d distribution for some d, even when the samples are of normal distributions.
- T in (1) is not a pivot, since its distribution depends on σ_X^2, σ_Y^2 (actually in terms of σ_X^2/σ_Y^2).
- If n and m both are large, we can resort to Central Limit Theorem as usual.

$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \overset{approx.}{\sim} N(0,1)$$

  The asymptotic approximation leads to a $(1-\alpha)100\%$ confidence interval for $\mu_X - \mu_Y$

$$\bar{X}_i - \bar{Y} \pm z_{1-\alpha/2}\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

  For example, a 95% asymptotic confidence interval for $\mu_X - \mu_Y$ is $\bar{X} - \bar{Y} \pm 1.96\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$.

- For small or moderate sample sizes n and m, there is a better approximation using t_ν distribution than the normal approximation.
$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \quad \sim \quad t_\nu \qquad \text{approximately}$$
with the degrees of freedom ν estimated by Welch-Satterthwaite approximation,
$$\nu \approx \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\left(\frac{S_X^2}{n}\right)^2/(n-1) + \left(\frac{S_Y^2}{m}\right)^2/(m-1)}$$
The asymptotic pivot leads to a $(1-\alpha)100\%$ confidence interval for $\mu_X - \mu_Y$
$$\bar{X}_i - \bar{Y} \pm t_{\nu,1-\alpha/2}\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$
with $\mathbb{P}(t_\nu \le t_{\nu,1-\alpha/2}) = 1 - \alpha/2$.

The Welch approximation works better when both n and m are not too small (say > 5), and when the two samples are of different sample sizes with substantially different sample variance.

Likelihood Ratio Test (Independent two normal samples with equal variance)

With distribution assumptions on sample data, maximum likelihood method estimates the parameters that make the observed data most probable, thus maximize the likelihood function. The actual maximum likelihood value is of little interest. It is the relative values of maximum likelihood functions under competing models that are compared to decide which hypothesis the data support.

Likelihood ratio test (LRT) is commonly used in statistical inference, especially with common probability distributions.

Likelihood ratio for hypotheses H_o, H_a with corresponding parameter space Θ_o and $\Theta_a = \Theta \setminus \Theta_o$ is defined as
$$LR = \frac{\max_{\Theta_o}\ell}{\max_\Theta \ell} = \frac{\text{maximum of the likelihood function when } H_0 \text{ is true}}{\text{maximum of the likelihood function over all}} \qquad \in \quad (0,1]$$
($\max_\Theta = \max_{\Theta_a \cup \Theta_o}$ is preferred to $\max_{\Theta_a}$ for technical reasons.) Smaller values of LR is evidence against H_o.

For large samples,
$$-2\log LR \sim \chi^2_d, \qquad d = dim\{\Theta\} - dim\{\Theta_o\}$$

Example — LRT to two independent normal samples with equal variance.

Suppoer $X_i, \cdots, X_n$ i.i.d. of distribution $N(\mu_X, \sigma^2)$, independent of $Y_1, \cdots, Y_m$ i.i.d. of $N(\mu_Y, \sigma^2)$.

Consider
$$\begin{cases} H_o: & \mu_X = \mu_Y \\ H_a: & \mu_X \ne \mu_Y \end{cases}$$

Given X_i $(i = 1, \cdots, n)$ and Y_j $(j = 1, \cdots, m)$, the joint likelihood is a function of the parameters given the observed data.
$$\ell = \ell(\mu_X, \mu_Y, \sigma^2 \big| X_1, \cdots, X_n, Y_1, \cdots, Y_m) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n+m} \prod_{i=1}^n e^{-\dfrac{(X_i - \mu_X)^2}{2\sigma^2}} \prod_{j=1}^m e^{-\dfrac{(Y_j - \mu_Y)^2}{2\sigma^2}}$$

The log-likelihood function is
$$L = \log \ell = -\frac{n+m}{2}\log(2\pi) - \frac{n+m}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \mu_X)^2 - \frac{1}{2\sigma^2}\sum_{j=1}^m (Y_j - \mu_Y)^2$$

We are going to find the set of parameters maximizing the likelihood function under each hypothesis.
Then the maximum likelihood values $\max_{\Theta_o}\ell$ and $\max_\Theta \ell = \max_{\Theta_a \cup \Theta_o}\ell$ are compared via their ratio.

There are various ways to derive the MLEs.

First, consider the formal way of obtaining MLEs using the smoothness of the likelihood function, taking partial derivations to find critical points as candidates, then verifying the maximum; an exercise and review of multivariate calculus, also matrix algebra.

MLE under H_o: $\mu_X = \mu_Y = \mu \in \mathbb{R}$, $\sigma^2 \in (0, \infty)$.

When the two samples are from the same distribution, we have a combined i.i.d. sample of size $n+m$.

Finding critical points of L where the derivatives are zero requires solving simultaneously the two equations
$$\begin{cases} \dfrac{\partial L}{\partial \mu} & = \dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^n (X_i - \mu) + \dfrac{1}{\sigma^2}\sum_{j=1}^m (Y_j - \mu) = 0 \\ \dfrac{\partial L}{\partial \sigma^2} & = -\dfrac{n+m}{2\sigma^2} + \dfrac{1}{2\sigma^4}\displaystyle\sum_{i=1}^n (X_i - \mu)^2 + \dfrac{1}{2\sigma^4}\sum_{j=1}^m (Y_j - \mu)^2 = 0 \end{cases} \qquad \mu \in \mathbb{R}, \quad \sigma^2 \in (0, \infty).$$

The solution set consists of one critical point,
$$\begin{cases} \hat{\mu}_o & = \dfrac{1}{n+m}\left(\displaystyle\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j\right) \\ \hat{\sigma}_o^2 & = \dfrac{1}{n+m}\left[\displaystyle\sum_{i=1}^n (X_i - \hat{\mu}_o)^2 + \sum_{j=1}^m (Y_j - \hat{\mu}_o)^2\right] \end{cases}$$

Recall for a function of one variable, if the function at a critical point of derivative zero has second derivative < 0, then the critical point is a local maximum of the function.

Here log-likelihood L is a function of two variables (μ, σ^2). For a critical point of L with partial derivatives $= 0$, the function L attains local maximum if its second derivative matrix,
$$H_L(\mu, \sigma^2) = \begin{bmatrix} \dfrac{\partial^2 L}{\partial \mu^2} & \dfrac{\partial^2 L}{\partial \mu \partial \sigma^2} \\ \dfrac{\partial^2 L}{\partial \sigma^2 \partial \mu} & \dfrac{\partial^2 L}{\partial (\sigma^2)^2} \end{bmatrix}$$
called **Hessian matrix**, is negative definite at the critical point.

To check negative-definiteness of the Hessian matrix at the unique point $(\hat{\mu}_o, \hat{\sigma}_o^2)$, it is sufficient to verify
$$\frac{\partial^2 L}{\partial \mu^2}\bigg|_{\hat{\mu}_o, \hat{\sigma}_o^2} < 0 \qquad \text{and} \qquad det\left[H_L(\hat{\mu}_o, \hat{\sigma}_o^2)\right] > 0$$
The negative-definiteness of a matrix is equivalent to the negativeness of the eigenvalues.

By the uniqueness of the critical point, the local maximum is global can be confirmed by comparing $L(\hat{\mu}_o, \hat{\sigma}_o^2)$ with the value of $L(\mu, \sigma^2)$ at the boundary of the parameter space, such as $\sigma^2 \to 0$ (exercise).

After the derivation and verifications, the MLEs under H_o are obtained as $(\hat{\mu}_o, \hat{\sigma}_o^2)$. Evaluating at the maximum $\ell(\hat{\mu}_o, \hat{\sigma}_o^2)$ and simplifying, the maximized likelihood under the null has a simple form, only in terms of σ_o^2, n, m.

$$\max_{H_o} \ell(\mu_X, \mu_Y, \sigma^2) \;=\; \ell(\hat{\mu}_o, \hat{\sigma}_o^2) \;=\; \left(\frac{1}{2\pi\hat{\sigma}_o^2}\right)^{(n+m)/2} e^{-(n+m)/2}$$

MLE under $H_a \cup H_o$: $(\mu_X, \mu_Y) \in \mathbb{R}^2,\ \sigma^2 \in (0, \infty)$.

Checking the critical points where the derivatives are zero by solving simultaneously the equation system

$$\begin{cases} \dfrac{\partial L}{\partial \mu_X} &= \dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{n}(X_i - \mu_X) = 0 \\ \dfrac{\partial L}{\partial \mu_Y} &= \dfrac{1}{\sigma^2}\displaystyle\sum_{j=1}^{m}(Y_j - \mu_Y) = 0 \\ \dfrac{\partial L}{\partial \sigma^2} &= -\dfrac{n+m}{2\sigma^2} + \dfrac{1}{2\sigma^4}\displaystyle\sum_{i=1}^{n}(X_i - \mu_X)^2 + \dfrac{1}{2\sigma^4}\displaystyle\sum_{j=1}^{m}(Y_j - \mu_Y)^2 = 0 \end{cases} \qquad (\mu_X, \mu_Y) \in \mathbb{R}^2, \quad \sigma^2 \in (0, \infty).$$

Method 1 for MLE under $H_a \cup H_o$ (outline)

Solving the three simultaneous equations, the solution set again consists of one critical point $(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$, with

$$\hat{\mu}_X = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}, \qquad \hat{\mu}_Y = \frac{1}{m}\sum_{j=1}^{m} Y_j = \bar{Y}, \qquad \hat{\sigma}_a^2 \;=\; \frac{1}{n+m}\left[\sum_{i=1}^{n}(X_i - \hat{\mu}_X)^2 + \sum_{j=1}^{m}(Y_j - \hat{\mu}_Y)^2\right]$$

Note that the usual unbiased estimator of σ^2 has denominator $n+m-2$ rather than $n+m$.

As the case for H_o, the unique solution is at a maximum of L can be checked by the negative-definiteness of the Hessian matrix

$$H_L(\mu_X, \mu_Y, \sigma^2) = \begin{bmatrix} \dfrac{\partial^2 L}{\partial \mu_X^2} & \dfrac{\partial^2 L}{\partial \mu_X \partial \mu_Y} & \dfrac{\partial^2 L}{\partial \mu_X \partial \sigma^2} \\ \dfrac{\partial^2 L}{\partial \mu_Y \partial \mu_X} & \dfrac{\partial^2 L}{\partial \mu_Y^2} & \dfrac{\partial^2 L}{\partial \mu_Y \partial \sigma^2} \\ \dfrac{\partial^2 L}{\partial \sigma^2 \partial \mu_X} & \dfrac{\partial^2 L}{\partial \sigma^2 \partial \mu_Y} & \dfrac{\partial^2 L}{\partial (\sigma^2)^2} \end{bmatrix}$$

at the unique critical point $(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$, then comparing with boundary limit points, etc. To check the negative-definiteness of the Hessian matrix at the unique point $(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$, it is sufficient to verify

$$\left.\frac{\partial^2 L}{\partial \mu_X^2}\right|_{(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)} < 0 \qquad \text{and} \qquad \left.\begin{vmatrix} \frac{\partial^2 L}{\partial \mu_X^2} & \frac{\partial^2 L}{\partial \mu_X \partial \mu_Y} \\ \frac{\partial^2 L}{\partial \mu_Y \partial \mu_X} & \frac{\partial^2 L}{\partial \mu_Y^2} \end{vmatrix}\right|_{(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)} > 0 \qquad \text{and} \qquad det\left[H_L(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)\right] < 0$$

Alternatively, the negative-definiteness of $H_L(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$ is equivalent to $\lambda_j < 0$ for all $j = 1, 2, 3$, where λ_j is the jth eigenvalue of $H_L(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$.

Method 2 for MLE under $H_a \cup H_o$

On the other hand, after solving the two easier equations of $\frac{\partial L}{\partial \mu_X} = 0$ and $\frac{\partial L}{\partial \mu_Y} = 0$, we can see that for any value σ^2, the maximizers $\hat{\mu}_X = \bar{X}$ and $\hat{\mu}_Y = \bar{Y}$ are unique and global. Therefore finding the maximum over all (μ_X, μ_Y, σ^2) is simplified to finding the maximum over σ^2 when $(\mu_X, \mu_Y) = (\hat{\mu}_X, \hat{\mu}_Y)$, which is a simpler, one-dimensional maximization problem.

$$\max_{H_a \cup H_o} \ell(\mu_X, \mu_Y, \sigma^2) = \max_{\mu_X, \mu_Y, \sigma^2} \ell(\mu_X, \mu_Y, \sigma^2) = \max_{\sigma^2} \ell(\hat{\mu}_X, \hat{\mu}_Y, \sigma^2)$$

To obtain maximizer of σ^2 of $L = \log \ell$ (thus of ℓ), set

$$\left.\frac{\partial L}{\partial \sigma^2}\right|_{(\hat{\mu}_X, \hat{\mu}_Y, \sigma^2)} \;=\; -\frac{m+n}{2\sigma^2} + \frac{\sum_{i=1}^{n}(X_i - \hat{\mu}_X)^2 + \sum_{j=1}^{m}(Y_j - \hat{\mu}_Y)^2}{2\sigma^4} = 0.$$

The unique critical point is the "pooled" estimate of the variance $\hat{\sigma}_a^2$. This is a maximum point of $L(\hat{\mu}_X, \hat{\mu}_Y, \sigma^2)$ since

$$\left.\frac{\partial^2 L}{\partial (\sigma^2)^2}\right|_{(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)} \;=\; \frac{m+n}{2\hat{\sigma}_a^4} - \frac{\sum_{i=1}^{n}(X_i - \hat{\mu}_X)^2 + \sum_{j=1}^{m}(Y_j - \hat{\mu}_Y)^2}{\hat{\sigma}_a^6} = -\frac{n+m}{2\hat{\sigma}_a^4} < 0.$$

Comparing with boundary limits verifies that the unique maximum $\hat{\sigma}_a^2$ is global, so is the globalness of $(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2)$.

The maximized likelihood can be evaluated and simplified as

$$\max_{H_a} \ell(\mu_X, \mu_Y, \sigma^2) = \max_{H_a \cup H_o} \ell(\mu_X, \mu_Y, \sigma^2) = \ell(\hat{\mu}_X, \hat{\mu}_Y, \hat{\sigma}_a^2) = \left(\frac{1}{2\pi\hat{\sigma}_a^2}\right)^{(n+m)/2} e^{-(n+m)/2}$$

The likelihood ratio has a simple form,

$$LR = \frac{\max_{H_o} \ell}{\max_{H_a \cup H_o} \ell} = \left(\frac{\hat{\sigma}_a^2}{\hat{\sigma}_o^2}\right)^{(n+m)/2}$$

By the standard asymptotic result for likelihood ratio (as covered in the previous course),

$$-2\log(LR) \quad \sim \quad \chi_d^2 \qquad under\ H_o$$

approximately, if $n+m$ are large, and the degree of freedom d is the difference of the dimensions of parameter space under $H_a \cup H_o$ and under H_0.

In the two independent, equal variance normal samples here,

$$-2\log(LR) = (n+m)\log\frac{\hat{\sigma}_o^2}{\hat{\sigma}_a^2}$$

The parameter space under $H_o : \mu_X = \mu_Y$ is

$$\{(\mu_X, \mu_Y, \sigma),\ \mu_X = \mu_Y \in \mathbb{R}, \sigma \in (0, \infty) = \mathbb{R}^+\} = \mathbb{R} \times \mathbb{R}^+$$

which is a 2-dimensional subspace $\subset \mathbb{R}^2$. The parameter space under $H_a : \mu_X \neq \mu_Y$ is

$$\{(\mu_X, \mu_Y, \sigma),\ \mu_X \in \mathbb{R}, \mu_Y \in \mathbb{R}, \sigma \in (0, \infty) = \mathbb{R}^+\} = \mathbb{R} \times \mathbb{R} \times \mathbb{R}^+$$

Therefore the degrees of freedom of the χ_d^2 is

$$d = dim(H_a \cup H_o) - dim(H_o) = dim(\mathbb{R}^2 \times \mathbb{R}^+) - dim(\mathbb{R} \times \mathbb{R}^+) = 3 - 2 = 1$$

Given data, critical values $\chi_{d,1-\alpha}^2$ with $\mathbb{P}(\chi_d^2 > \chi_{d,1-\alpha}^2) = \alpha$ can be used to conduct hypothesis testing and to construct confidence intervals. For example, $\chi_{1,0.95}^2 = 3.84$.